```
from os import name
#practice tasks
# functions

def func():
  print("Hello from function!")
func()
print("\n")


def func_name(fname):
  print(fname + " Welcome")
func_name("Ayesha")
print("\n")


def details(lname,lemail):
  print("Name: ",lname, "Email: ",lemail)
details("Ayesha", "a@gmail.com")
print("\n")


def country(cname):
  print("I am from ",cname)
country("Pakistan")
print("\n")


def child(child1,child2,child3):
  print("Youngest child is ",child3)
child(child1="Ali", child2="Sara", child3="Amna")
print("\n")


def func_food(food):
  for x in food:
    print(x)

fruits=["Mango", "Strawberry","Kiwi"]
func_food(fruits)
print("\n")


def func_pass():
  pass



#class

class MyClass:
  x = 5
p1 = MyClass()
print(p1.x)
print("\n")


class Person:
  def __init__(self, name, age):
    self.name=name
    self.age=age

p1 = Person("Ayesha",22)
print("Name: " ,p1.name)
print("Age: ",p1.age)
print("\n")

class person:
  def __init__(self, name,age):
    self.name=name
    self.age=age
  def myfunc(self):
   print("Hello my name is "+self.name)

p2=person("Ayesha",22)
```

```
p2.myfunc()

del p2.age
del p1
del p2

class person:
  pass
```

```
Hello from function!


Ayesha Welcome


Name:  Ayesha Email:  a@gmail.com


I am from  Pakistan


Youngest child is  Amna


Mango
Strawberry
Kiwi


5


Name:  Ayesha
Age:  22


Hello my name is Ayesha
```

```
#tasks
#task 1

roll_no = [1, 5, 6, 7, 8, 10]

def attendance(roll_number):
    if roll_number in roll_no:
        return "Present"
    else:
        return "Absent"

print(attendance(1))
```

```
Present
```

```
#task 2
class fruit:
  def __init__(self, name, color):
    self.name=name
    self.color=color
f1=fruit("Apple","Red")
f2=fruit("Cherry","Red")

print("Fruit Name  Fruit Color")
print(f1.name,"  ",f1.color)
print(f2.name, " ",f2.color)

f1.color="Green"
print(f1.name, " ",f1.color)
```

```
Fruit Name  Fruit Color
Apple    Red
```

```
Cherry   Red
Apple   Green
```

```
#task 3
class Student:
  def __init__(self,name,age,grades):
    self.name=name
    self.age=age
    self.grades=grades
  def avg(self):
    return sum(self.grades)/len(self.grades)

s1=Student("Max", 22,[10,10,10])
s2=Student("peter",21,[9,9,8])

print("Name:", s1.name)
print("Average Grade:", s1.avg())
print("\n")
print("Name:", s2.name)
print("Average Grade:", s2.avg())
```

```
Name: Max
Average Grade: 10.0


Name: peter
Average Grade: 8.666666666666666
```

```
#task 4
class Employee:
    def __init__(self, name, salary):
        self.name = name
        self.salary = salary

    def display_details(self):
        print("Name:", self.name)
        print("Salary:", self.salary)


class Manager(Employee):
    def __init__(self, name, salary, department):
        super().__init__(name, salary)
        self.department = department

    def display_details(self):
        print("Manager Name:", self.name)
        print("Salary:", self.salary)
        print("Department:", self.department)


class Developer(Employee):
    def __init__(self, name, salary, programming_language):
        super().__init__(name, salary)
        self.programming_language = programming_language

    def display_details(self):
        print("Developer Name:", self.name)
        print("Salary:", self.salary)
        print("Programming Language:", self.programming_language)



m1 = Manager("Ayesha", 90000, "HR")
d1 = Developer("Ali", 80000, "Python")


m1.display_details()
print()
d1.display_details()
```

```
Manager Name: Ayesha
Salary: 90000
Department: HR

Developer Name: Ali
Salary: 80000
Programming Language: Python
```

```
#task 5

class Shape:
    def area(self):
        pass


class Circle(Shape):
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        return math.pi * self.radius ** 2


class Rectangle(Shape):
    def __init__(self, length, width):
        self.length = length
        self.width = width

    def area(self):
        return self.length * self.width


class Triangle(Shape):
    def __init__(self, base, height):
        self.base = base
        self.height = height

    def area(self):
        return 0.5 * self.base * self.height



shapes = [
    Circle(7),
    Rectangle(10, 5),
    Triangle(6, 4)
]

for shape in shapes:
    print("Area:", shape.area())
```

```
Area: 153.93804002589985
Area: 50
Area: 12.0
```